

```

1 class RuleBasedSystem:
2     def __init__(self, facts, rules):
3         self.facts = set(facts)
4         self.rules = rules
5
6     def forward_chain(self):
7         iterations = 0
8
9         while True:
10             new_fact_added = False
11
12             for rule in self.rules:
13                 # Check if all conditions of the rule exist in current facts
14                 if all(cond in self.facts for cond in rule['if']):
15                     if rule['then'] not in self.facts:
16                         print(
17                             f"Rule Triggered: IF {rule['if']} "
18                             f"THEN Add {rule['then']}"
19                         )
20
21                         self.facts.add(rule['then'])
22                         new_fact_added = True
23                         iterations += 1
24
25             # Stop if no new fact was added

```

```
        if not new_fact_added:
            break

    return self.facts
```

```
# Example Usage
```

```
if __name__ == "__main__":
```

```
    # Student Details
```

```
    name = "Zainab Khatoon"
```

```
    reg_no = "24BECS173"
```

```
    # Base facts
```

```
    initial_facts = [
```

```
        'Socrates_is_human',
```

```
        'All_humans_are_mortal'
```

```
    ]
```

```
    # Production rules
```

```
    production_rules = [
```

```
        {
```

```
# Production rules
production_rules = [
    {
        'if': [
            'Socrates_is_human',
            'All_humans_are_mortal'
        ],
        'then': 'Socrates_is_mortal'
    }
]

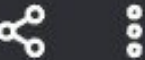
# Create Rule-Based System
rbs = RuleBasedSystem(initial_facts, production_rules)

# Display Student Details
print("=====")
print("Name    :", name)
print("Reg No :", reg_no)
print("=====")

print("Initial Facts:", initial_facts)

# Perform forward chaining
final_kb = rbs.forward_chain()
```

Output



```
=====
```

```
Name    : Zainab Khatoon
```

```
Reg No  : 24BECS173
```

```
=====
```

```
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
```

```
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_m
```

```
Final Knowledge Base Facts: {'Socrates_is_mortal', 'All_humans_are_mortal',  
'Socrates_is_human'}
```

```
=== Code Execution Successful ===
```